

STATEMENT OF WORK (SOW)

George Washington Birthplace National Monument

Repoint Brick Exterior and Install Copper Gutters

BACKGROUND AND PURPOSE

George Washington Birthplace National Monument is located in the Northern Neck of Virginia. It encompasses 551 acres of land where seven generations of the Washington family lived and where George Washington was born in 1732. The park was also central to one of the earliest efforts to memorialize George Washington during the celebration of his bicentennial birth anniversary in 1932.

The Memorial House Museum (MHM) sits at the same location of the obelisk monument the War Department had placed in 1896. The construction of the MHM was completed in 1931 and on May 14, 1932, the MHM was officially opened by the National Park Service. The Colonial Revival Kitchen (CRK) was constructed in 1931 and sits adjacent to the MHM.

This project will repair deteriorating mortar joints on the exterior brick surface of the Memorial House and Colonial Kitchen buildings. Half round gutters and downspouts will be installed on the MHM and CRK.

GENERAL INFORMATION

1. This contract is a Fixed-Firm Price contract for repointing the brick mortar joints and the installation of new half round copper gutters and downspouts on two (2) buildings.
2. The contractor will analyze mortar samples on both buildings to determine proportions of constituent parts and to develop a mortar mix that will match the most prevalent mortar in color and texture. Samples of recommended mortar will be provided to the COR for review and approval. Mortar joint mockups shall be provided to the COR for approval to determine the proper color, texture, and joint pattern.
3. Work consists of cleaning and removing the biological growth on the brick surface, repointing the mortar joints on the exterior walls and chimneys. All exterior brick wall sections will be inspected for deficiencies. The primary focus will be to repair decayed mortar joints. Damaged, fractured or broken brick replacements will be evaluated with COR prior to removal and replacement. The contractor will inspect the chimney mortar caps on six (6) chimneys and replace the mortar caps to prevent water infiltration.
4. The contractor will install new seamless half round copper gutters and downspouts. Including all hangers, fasteners, end caps, straps and associated components for the completed installation the gutters on both buildings.
5. The work will require the contractor to maintain public access to both buildings during the on-going work. Work will also include daily site clean-up and disposal of debris.
6. **Location:** George Washington Birthplace NM 1732 Popes Creek Road Colonial Beach, VA 22443.

7. **Current Conditions:**

- The MHM exterior brick surface: Approximately 2,600 square feet.
- The CRK exterior brick surface: Approximately 400 square feet.

8. **Regular Working Hours:** Work is to be performed between 7am and 3:30pm, Monday through Friday. No work is to be performed on Saturday or Sunday or federally recognized holidays (unless otherwise authorized by the Contracting Officer's Representative 72 Hours ahead of the date of work).

9. **Period of Performance:** 60 calendar days after Notice to Proceed is issued.

10. **Notification:** The Contractor shall notify the Contracting Officer (CO) and the Contracting Officers Representative (COR) who serves as the single point of contact, at least 48 hours in advance for all work in park facilities.

11. **Waste Disposal:** The Contractor shall properly dispose of solid waste materials offsite.

12. **Park:** George Washington Birthplace National Monument (GEWA).

PRE-BID SITE EXAMINATION

1. **Site Visit:**

- All prospective Bidders, and their Subs, as appropriate are strongly encouraged to attend the pre-bid site visit and examine the existing conditions of the Project prior to submission of Bid in order to better understand the Work, and aid in forming any questions prior to the submission of Bid.
- Any and all discrepancies shall be forwarded to the Contracting Officer in writing for clarification prior to the commencement of Work. Any Work performed prior to said clarification, particularly and including that which may require correction shall be corrected by the Contractor with no additional costs to the Contract and Government and no time extension shall be made to the Contract Schedule. Lack of knowledge of the existing conditions will not be allowed as a basis for extra compensation.

DESCRIPTION OF WORK

1. The contractor shall provide all labor, equipment, material, and supervision for repointing the mortar joints and the installation of new gutters and downspouts on two (2) buildings at GEWA.
2. The contractor shall provide a mortar analysis of the existing mortar on both buildings to determine proportions of constituent parts and to develop a mortar mix that will match the most prevalent mortar in color and texture.
3. The contractor shall provide a mockup of the proposed mortar joint for COR approval.
4. The contractor shall use D/2 Biological Solution to clean and remove all biological growth on the brick surfaces of both buildings prior to repointing the mortar joints.
 - D/2 Biological Solution shall be used according to the manufacturer's instructions.

- Protect all existing adjacent materials such as doors, windows, flashings, roofing, plants and other existing materials that are not intended to be treated. Protect areas and surfaces not being treated with barriers suitable for the product being used. Appropriate care should be taken at air intakes, air conditioning vents and similar openings that may come in contact with the product. Take appropriate precautions to avoid harm to building occupants, pedestrians and nearby property. For any number of reasons it is essential to maintain a high degree of worker and occupant safety while working with biological solution.
- Allowable Equipment to be used for cleaning the surfaces of the buildings are natural bristle brushes, soft, clean rags, low-pressure applicator, such as pump sprayer or battery powered sprayer.
- Pressure washers are not allowed.

5. The contractor shall remove loose and deteriorated mortar joints.

- Prepare the work area prior to starting any work on the exterior brick surfaces. Begin by protecting adjacent surfaces, landscaping, and building components with drop cloths or other suitable barriers. All debris should be contained and disposed of daily. Ensure all necessary safety equipment is in place for workers.
- Assess the mortar joints, carefully inspect the brickwork to identify areas with deteriorated, loose, or cracked mortar. Mark these sections to guide removal efforts and avoid disturbing sound joints.
- The use of power tools, grinders, or impact tools that could damage the historic bricks or surrounding material are not permitted. Unless the contractor provides a proven method that will not cause any damage or harm the historic fabric of the building. Use only hand tools such as masonry chisels, joint rakes, and small hammers.
- Clean the joints after removing the old mortar, brush out the joints using a stiff-bristled brush and compressed air to remove dust and debris. Ensure the joint is clean to a depth of 1/2 to 1 inch, or as specified for repointing.
- Inspect the area for damage, examine the exposed bricks and joint cavities for any additional damage. If bricks are loose or broken, consult with the COR for evaluation before proceeding with any replacement.

6. The contractor shall repoint the mortar joints using an approved Lime Based Mortar.

- Existing Mortar Mix:
 - Lime putty and fine yellow "front sand"
 - Not over 10-percent of white "Medusa" cement
- The joints will then be repointed in lifts to ensure a proper curing time. Using a pointing trowel, pack the mortar firmly into the joints in layers, ensuring full contact with the brick edges. Fill the joint completely, avoiding voids or air pockets. Once the joints have been pointed, they will be struck to match the historic mortar joints and brush finished.

- Allow the mortar to cure, let the new mortar cure slowly, protecting it from direct sun, wind, or heavy rain. Mist the repointed area with water as needed to prevent premature drying and cracking.
- Clean and remove mortar residue once the mortar has fully cured, gently clean any mortar residue from the brick surface using a soft brush and clean water. Avoid harsh scrubbing or acidic cleaners to protect the historic brickwork.

7. The contractor shall install 6-inch half round seamless copper gutters and downspouts.

- Measure and ensure all gutter lengths, downspout locations, and attachment methods match specifications for both the MHM and CRK.
- Mark the location of the roof's rafter tails. Make every effort to position and secure the gutter brackets/hangers at the rafter tail marks, spacing between hangers must be no more than 24-inches apart for proper support.
- Secure gutter brackets/hangers to the building using 1-1/2" stainless steel or copper screws.
- Provide and install 20 oz. copper 6-inch reverse bead half round seamless gutters along all eaves of both buildings. The high point for the new gutters will be the center and pitched down to drain at both ends of the gutter.
- Provide and install 20 oz. copper 3-inch plain round downspouts at both ends of building.
- Ensure gutters protrude 6 inches from the ends of the roof eaves, allowing downspouts to drain on the short sides of the buildings.
- Attach downspout outlets to the gutter sections using copper outlet tubes, ensuring a watertight fit. Connect downspouts to the outlets, securing them with copper straps anchored into mortar joints.
- All joints and connections shall be soldered with 50/50 Mix - Tin/Lead to prevent leaks.
- Test the system by running water through the gutters to ensure proper flow toward downspouts and splash pads, confirming that water is directed away from the foundation and sidewalks.

8. Scaffolding, Ladders and Manlifts:

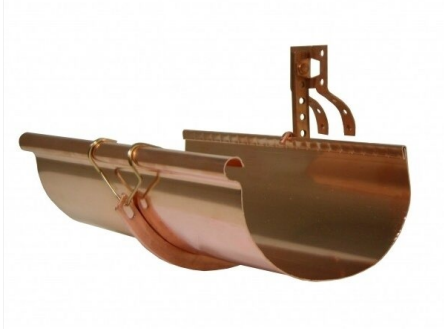
- Scaffolding and ladders are allowed to be used by contractor.
- Ensure that the scaffolding at the job site is properly set up in accordance with OSHA regulations.
- Scaffolding and ladders must be inspected for stability and proper assembly, and perimeter barriers should be installed to prevent unauthorized access to the work area.

Additionally, clear signage should be posted to alert others of ongoing construction activities and any potential

- Manlifts or JLG lifts may be used but only with the approval of the COR. If such lifts are used by the contractor, heavy duty ground protection mats shall be placed on the ground to prevent damage or rutting of the turf/grass. The manlift equipment must remain on the protective mats at all times when on site.

PRODUCTS

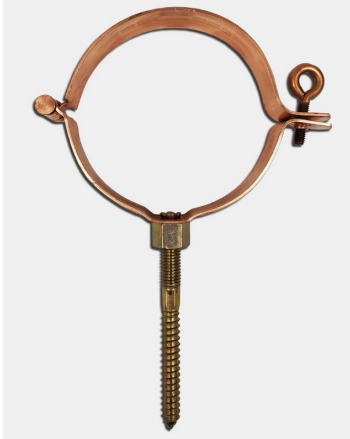
1. D/2 Biological Solution
2. Lime-Base Mortar
3. Reverse Bead Copper Half Round Gutter, 20 oz. copper



4. Plain Round Copper Downspout, 20 oz. copper



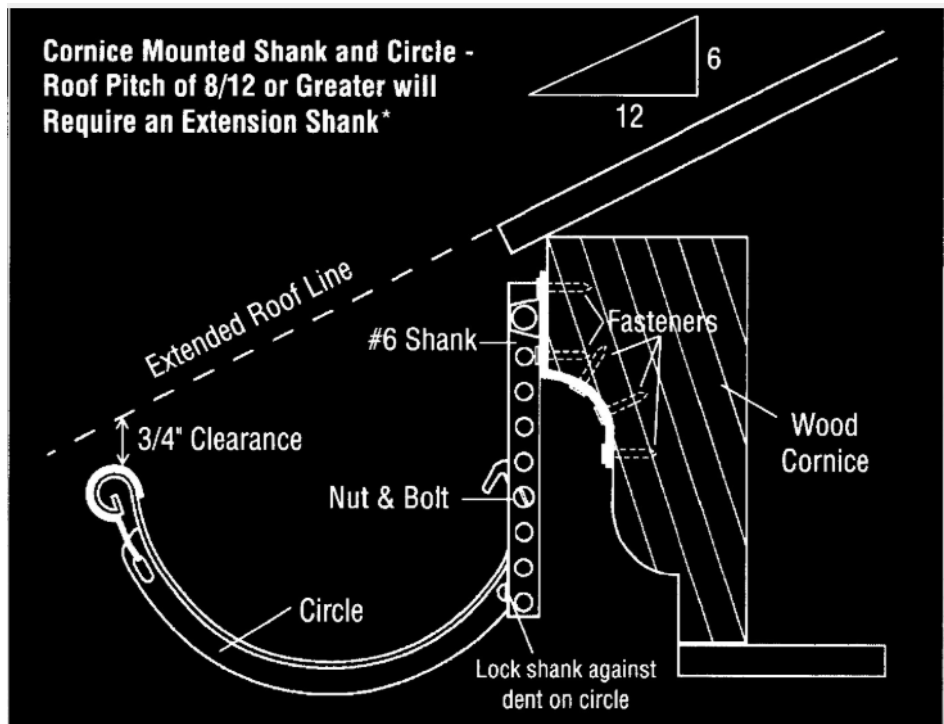
5. Heavy Duty Stamped Plain Round Copper Downspout Bracket



6. Crown Molding Copper Shank #6 for Half Round Gutter Hanger



7. Copper Gem Circle Half Round Gutter hanger



PROTECTION AND PREVENTION

1. Park Facilities:
 - Protect Park facilities and surrounding area from damage and/or contamination due to construction activities at the jobsite. Contractor shall implement procedures to ensure no damage is caused by workers and/or construction related activities to the buildings and surrounding area.
2. Housekeeping:
 - The Contractor shall:
 - Keep the project neat, orderly, and in safe condition at all times. Dispose of debris daily.
 - Provide enough containers for collecting construction debris and construction materials to be recycled.
 - Keep volatile waste in covered containers

DISPOSAL

1. Demolition/Removal:
 - All related construction debris generated from the removal or installation process must be collected and disposed of properly in accordance with applicable Federal and State regulations.

SAFETY

1. Accident Prevention Plan:
 - After contract award and before the Pre-Construction conference, submit for review an Accident Prevention Plan.
2. First Aid:
 - Always keep an updated first aid kit on site.
3. Personnel Protective Equipment:
 - Meet requirements of applicable ANSI standards.
 - All workers must be provided with appropriate personal protective equipment (PPE), such as hard hats, safety glasses, gloves, and dust masks.
4. Emergency:
 - In case of an emergency call 911 and then notify the COR.

QUALITY ASSURANCE AND EXPERIENCE

1. The contractor shall establish and maintain a complete Quality Control Plan acceptable to the Government to ensure the requirements of the contract are provided as specified. The Government shall inspect completed units as a basis for acceptance and payments and provide recommendations to improve work quality while work is in progress. The Contractor is responsible for providing quality control to assure that work complies with requirements of contract specifications.

2. **Experience:**

- All prospective bidders must provide evidence of 5-years of past work experience in the masonry trade and the installation of gutters and downspouts along with associated components.

CONDUCT

1. Contractor workers/technicians shall always present themselves in a professional manner.
2. No cursing or inappropriate language shall be used on the jobsite.
3. No smoking in park owned facilities. Smoking area will be designated by the COR.
4. Cigarettes shall be extinguished and placed in approve receptacle for disposal.
5. No drinking or drug use is allowed on the jobsite.

STAGING AND STORAGE

1. **Staging:**

- There are limited areas for staging on site. The contractor will work with the identified COR to define a "Limits of Work" and "Staging Area".

8. **Storage:**

- The contractor will take full responsibility for any tools, equipment and materials stored on site. The park is not responsible for anything stored on site.

SUBMITTALS

1. **Submittals:**

- The following items shall be submitted a minimum of 10 calendar days prior to the preconstruction conference.
 - Construction Schedule
 - Breakdown of the Schedule of Values
 - Accident Prevention/Safety Plan
 - Letter designation Project Superintendent
 - List of Subcontractors prior to notice to proceed
 - Product and Materials

WARRANTY

The contractor will provide a one-year on-call basis warranty for all work performed. The contractor will provide a written warranty stating that work executed under this contract will be free from defects of materials and workmanship for a period of two years from date of substantial completion; any equipment breakdown, leaks, electrical issues, etc. will be remedied on written notice at no additional cost to the government. The warranty shall be in writing and shall be signed by the contractor. The warranty shall include the removal and replacement of defective equipment and materials as required at no cost to the government.

PHOTOS

MHM Back



MHM Front



CRK

